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REVIEW ARTICLE



Flexible, ultrathin bioelectronic materials and devices for chronically stable neural interfaces

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Abstract

Advanced technologies that can establish intimate, long-lived functional interfaces with neural systems have attracted increasing interest due to their wide-ranging applications in neuroscience, bioelectronic medicine, and the associated treatment of neurodegenerative diseases. A critical challenge of significance remains in the development of electronic platforms that offer conformal contact with soft brain tissue for the sensing or stimulation of brain activities and chronically stable operation in vivo, at scales that range from cellular-level resolution to macroscopic areas. This review summarizes recent advances in this field, with an emphasis on the use of demonstrated concepts, constituent materials, engineered designs, and system integration to address the current challenges. The article begins with an overview of recent bioelectronic platforms with unique form factors, ranging from filamentary probes to conformal sheets and three-dimensional frameworks for alleviating the mechanical mismatch between interface materials and neural tissues. Next, active interfaces which utilize inorganic/organic semiconductor-enabled devices are reviewed, highlighting various working principles of recording mechanisms including capacitively and conductively coupled sensing enabled by high transistor matrices at high spatiotemporal resolution. The subsequent section presents approaches to biological integration which use active materials for multiplexed addressing, local amplification and multimodal operation with high-channel-count and large-scale electronic systems in a safe fashion that provides multi-decade stable performance in both animal models and human

Lianjie Zhou and Zhongyuan Wu contributed equally to this work.

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subjects. The advances summarized in this review will guide the future direction of this technology and provide a basis for next-generation chronic neural interfaces with long-lived high-performance operation.

KEYWORDS

neural interface, flexibility, passive electrodes, active recording, barrier materials, chronic stability

1 | INTRODUCTION

Promising and attractive applications of a neural interface range from neuroscience research,^[1,2] brain-computer interfacing (BCI),^[3,4] and body function reconstruction^[5,6] to the diagnosis and treatment of neurological disorders.^[7–9] Substantial efforts have been devoted to developing such platforms to facilitate the recording and modulation of the activity of neural systems, either at the level of individual neurons or across large areas and volumes of the brain.^[10–15] Non-invasive devices, such as wearable electroencephalogram (EEG) arrays, can measure brain activity at a distance from the neural tissue with little risk, but suffer from signal distortion, low spatial resolution, and background noise due to the attenuating and low-pass filtering effect of the skull and scalp.^[16,17] On the other hand, implantable neural interfaces which use various forms of non-penetrating flexible electrode arrays and penetrating probes, although they operate in an invasive manner, are capable of recording the electrocorticography (ECoG) of the cerebral cortex and the local field potential (LFP) inside the brain tissue with much better signal quality and higher spatial resolution.^[18,19]

The development of representative implantable neural interfaces over the last 50 years is summarized in Figure 1. Materials typically used in conventional platforms, such as Si, SiO₂, and metal electrodes, exhibit high rigidity, with Young's modulus for these materials being 8–9 orders of magnitude higher than that of neural tissue.^[31] Advances in materials and microfabrication techniques have facilitated the emergence of implantable interfaces with integrated functions on flexible supporting materials like polydimethylsiloxane (PDMS), parylene C, and polyimide (PI).^[32–36] In passive neural recording arrays, the electrical potential is directly measured using a metal electrode that contacts the brain tissue and is transmitted to an external system for further processing. The difficulties in wiring and interconnect routing become significant with the increase of channel counts.^[37] As a result, it is challenging to develop large-area and high spatial resolution interfaces with passive electrode arrays. Alternatively, active neural recording using inorganic materials that are compatible with the fabrication schemes in the semiconductor industry allows the miniaturization and integration of large-scale electrode arrays. The introduction of transistors into active arrays provides in situ signal amplification capability and multiplexed addressing, making it possible to perform high-quality and high-resolution mapping of brain signals. High channel-density

Key points

What is already known about this topic?

- Substantial efforts have been devoted to developing platforms for recording and modulating the activity of neural systems, either at the level of individual neurons or across large areas and volumes of the brain. However, challenges remain in terms of avoiding performance degradation and device failure due to the reaction to a foreign body implanted in the body environment.

What does this study add?

- The article reviews recent bioelectronic platforms with unique form factors, ranging from filamentary probes to conformal sheets and three-dimensional frameworks, for alleviating the mechanical mismatch between interface materials and neural tissues. Additional discussions emphasize their integration with actively multiplexed, amplified bioelectronic systems for electrophysiological mapping and stimulation with high spatial-temporal resolution and chronic stability.

micro-electrocorticography (μ -ECoG) arrays with signal amplifier transistors and multiplexing transistors are a state-of-the-art example of this type of active interface.^[24] Although most current neural interfaces are optimized in terms of materials and structures to minimize the damage to the brain, the wires and connectors in tethered devices protruding through the skull to external hardware inevitably increase infection risk and constrain objects' free movement. To this end, tether-free and fully wireless systems that integrate miniaturized designs and advanced power transfer technologies have been developed.^[27,38]

Neural interfaces using biodegradable materials for short-term applications and using encapsulation materials for long-term applications are two important directions in the development of implantable platforms.^[39–44] Compared with acute signal recording, platforms which enable chronically stable operation have important applications in the study of long-term learning, memory formation processes, and neural disease treatment.^[45] However, challenges remain in terms of

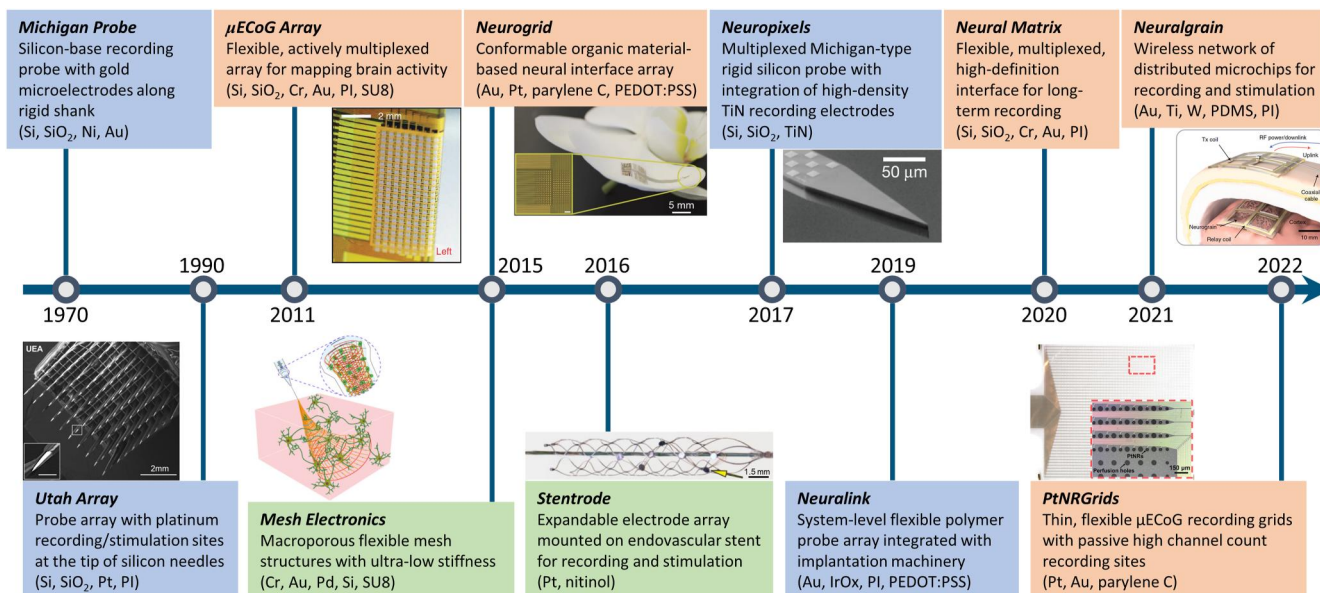


FIGURE 1 Development of representative neural interfaces. The key features and adopted materials of the platforms displayed in each channel are summarized. The different colors in the panel categorize the neural interfaces into three different form factors. Needle-shaped probe: Michigan Probe,^[20] Utah Array,^[21] American Chemical Society, Neuropixels,^[22] AAAS, and Neuralink^[23]; Flexible-sheet structure: μECoG Array,^[24] Springer Nature Ltd, Neurogrid,^[25] Springer Nature Ltd, Neural Matrix,^[26] Neuralgrain,^[27] Springer Nature Ltd, and PtNRGrids,^[28] AAAS; Three-dimensional structure: Mesh Electronics,^[29] Springer Nature Ltd and Stentrode,^[30] Springer Nature Ltd.

avoiding performance degradation and device failure due to the reaction of the implanted body environment to foreign bodies.^[46] One of the key issues which needs to be addressed is the mechanical mismatch between brain implants and brain tissue. Considering the soft and vulnerable properties of neural tissue (Young's modulus in the range of 0.1–16 kPa),^[47] specific features including low stiffness to minimize brain damage induced by device displacement after implantation, and good biocompatibility to reduce the foreign-body immune response are required for the chronic operation of the implantable interfaces. In addition, thin and highly flexible forms of encapsulation layers are required to prevent biofluids from penetrating into the device; this has been difficult to achieve using polymer-based materials. The selection of effective barrier materials and their integration with implantable active platforms represents an attractive research area.^[48]

This review summarizes recent progress in these directions with an emphasis on the engineered materials, device designs, and integration schemes of neural interfaces for brain signal recording and stimulation that offer long-term, stable, high-channel-count, and high-resolution operation. Platforms which provide stable performance within the intended period (depending on the actual application scenario) are also involved in this discussion. The article begins with a historical overview of various platforms with form factors ranging from flexible filaments to conformal sheet-like mechanics and complex three-dimensional structures. Subsequent sections review different classes of active flexible materials for these purposes, with comparative evaluations of diverse types of designs and recording or stimulation strategies, highlighting the use of silicon-based complementary metal-oxide-semiconductor (CMOS) technology. Additional discussions

emphasize their integration with actively multiplexed, amplified bioelectronic systems which facilitate electrophysiological mapping and stimulation with high spatial-temporal resolution and chronic stability in operation, whereby systems can incorporate tens of thousands of active pixels distributed in variable densities at full, brain-scale dimensions over a period of decades in vivo. Insights gained from the use of the materials and devices summarized in this article will inspire continuing advancements in the long-term stability of functions and benign interfaces with biological systems, with the goal of achieving various therapeutic and diagnostic applications, ranging from animal model studies to human healthcare.

2 | CLASSES OF ELECTRONIC PLATFORMS WITH PASSIVE ELECTRODES

Chronic and stable bioelectronic implants are becoming increasingly important in biomedical research and clinical medicine. Such implants are designed to be placed inside the body and interact with biological systems, allowing real-time monitoring of physiological signals and targeted delivery of electrical or chemical stimulation.^[11,49,50] Although the commonly used materials still have relatively high Young's moduli, as shown in Table 1, mechanically compliant interfaces in the form of flexible probes, thin-sheet structures, or mesh structures have been developed through the structural design of polymer and elastomer materials with ultra-thin thickness; these provide significantly lower stiffness than bulkier materials. The mechanism can be explained by the negative dependence of bending stiffness on probe

TABLE 1 Functions, mechanical properties, and electrical properties of materials commonly used in neural interfaces.

Materials	Function	Young's modulus (GPa)	Materials	Function	Electrode impedance at 1 kHz ($M\Omega \mu m^2$)	Charge storage capacity ($mC cm^{-2}$)
Si	Substrate	130 ^[51]	Au	Electrode	232	0.4 ^[52]
SiO ₂	Encapsulation	70 ^[51]	Pt	Electrode	250	0.35–0.5 ^[53]
Polyimide	Substate/Encapsulation	4 ^[24]	IrOx	Electrode	40	4 ^[54]
Paralene C	Substate/Encapsulation	2.7 ^[25]	TiN	Electrode	40	0.87 ^[54]
PDMS	Substate	3×10^{-4} – 1×10^{-3} ^[55]	Graphene	Electrode	1500	1.9 ^[56]
SU-8	Substate/Encapsulation	3 ^[57]	PtIr	Electrode	135	0.13 ^[58]
Silk fibroin	Coating	4–6 ^[59]	PEDOT:PSS	Electrode	23.6	2.3 ^[60]

thickness, in the form of $K = E \times w \times h/12$, where E is the Young's modulus of constituent materials, and h and w are the thickness and width of the filaments, respectively.^[61] An example of a neural interface which utilizes this strategy, which will be discussed in detail later in this review, is neuro-like mesh electronics. This mimics the three-dimensional network structure of neurons with over 90% macroporosity, exhibiting comparable stiffness to that of brain tissue and provoking a negligible immune response.^[29] Figure 2 shows recent examples of needle-shaped probes (Figure 2A), platforms with conformal-sheet design (Figure 2B), and three-dimensional complex structure (Figure 2C). The mechanical constructs, shapes, and layouts were designed to minimize tissue damage during implantation and in vivo operation, providing a secure and stable interface with the target biological tissue.

2.1 | Penetrating needle-shaped filamentary probes

The Neuralink device shown in Figure 2A-I has recently attracted the attention of scholars.^[23] The recording array integrates 3072 electrodes on 96 threads (24 μm in diameter), which are packaged into a small implantable device that contains a custom chip of $23 \times 18.5 \times 2$ mm³, with a neurosurgical robot specifically designed for implantation. Broadband recording through more than 3000 electrodes inserted into a free-moving rat was achieved. The results show that Neuralink has unprecedented packaging density and scalability in a clinically relevant package.

Neuron-like electronics (NeuE), which organize neuron-like building blocks to construct an open three-dimensional electronic network, is also an excellent work of biomimetic electronics in a needle-shaped probe array (Figure 2A-II).^[62] NeuE consists of a polymer/metal/polymer structure, and the total thickness of the probe is about 0.9 μm . The bending stiffness of the NeuE's neurite-like interconnect is comparable to that of the axon of a neuron, and the estimated percentage of the occupied brain volume of the implanted NeuE probe is 0.07%–0.3%, 20 times lower than that of recent macroporous mesh probes.^[68] Although

not yet fully functional, NeuE is a successful case of the use of bioinspiration and biomimicry as a tool for designing innovative materials and devices for neurotechnology.

Long-term, stable electrophysiological recording of thousands of neurons in the brain is one of the main purposes of electrodes in the field of neurotechnology.^[69] Figure 2A-III shows ultra-flexible nanoelectronic thread electrodes (NETs) with a cross-sectional area of only 30–100 μm^2 ; these have been used for the stable electrophysiological recording of thousands of neurons at a density of about 1000 neural units per cubic millimeter for several months in head-fixed mice and free-moving rats.^[63]

2.2 | Conformal intimately coupled sheets

Compared with needle-shaped probes that need to be precisely implanted into the brain, the small size and high-density configuration of conformal-sheet design in μ -ECoG arrays allow for the precise localization and characterization of neural activity with minimal invasiveness.^[70] Figure 2B-I illustrates an organic-material-based, ultra-conformable, biocompatible, and scalable neural interface array (NeuroGrid) that can record the local field potentials and action potentials of cortical surface neurons.^[25] The electrode array has a $10 \times 10 \mu m^2$ electrode surface area and 30 μm inter-electrode spacing, which matches the average size of neuronal bodies and neuronal density. The device uses poly (3,4-ethylene dioxythiophene) doped with poly (styrene sulfonate) (PEDOT:PSS) as the interface material to reduce the electrochemical impedance mismatch between the tissue and the electrodes. The incorporation of multiple piercings into the NeuroGrid structure allows the free flow of cerebrospinal fluid and minimizes disruption during chronic recording. Continuous observation of the lasting recording of multiple neurons in rats (for 10 days) meets the typical duration of diagnostic recording in patients with epilepsy.

The interfacial impedance between the metal electrode array and tissue has a negative relationship with electrode size. It has been reported that electrically conductive hydrogels (ECHs, Figure 2B-II) can maintain high elasticity (stretchable up to 200%) and low modulus (<30 kPa) while

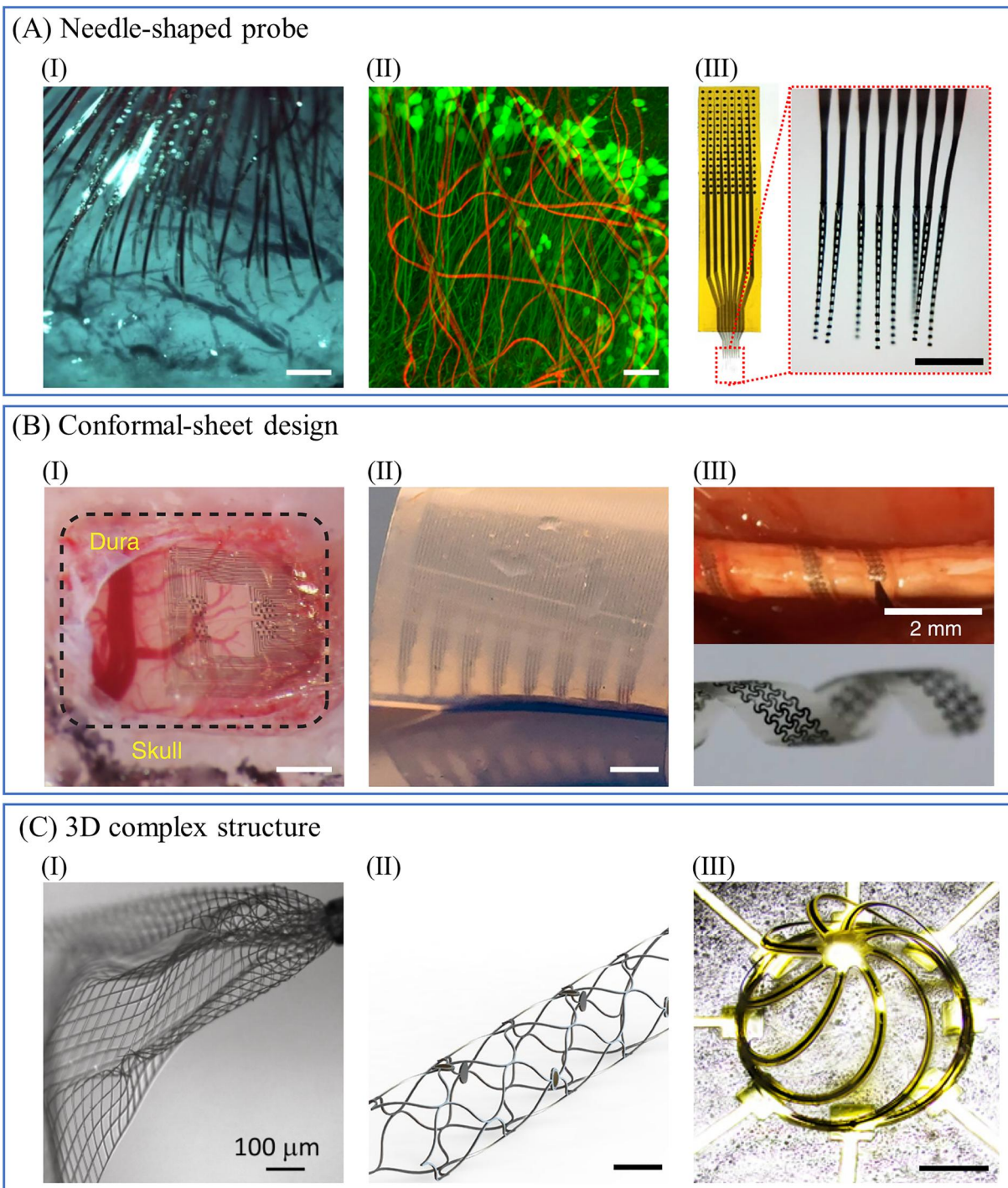


FIGURE 2 Novel implantable neural interfaces for biomedical applications. (A) Needle-shaped probe. (A-I) Flexible passive probe array with gold recording electrodes packaged by polyimide substrates (probe size of $20\text{ mm} \times 5\text{--}50\text{ }\mu\text{m} \times 4\text{--}6\text{ }\mu\text{m}$). Scale bar: $500\text{ }\mu\text{m}$. (A-II) Dendrite interconnects consist of a SU-8/Pt/SU-8 structure with passive electrodes (platinum, circular area $\sim 200\text{ }\mu\text{m}^2$, thickness 50 nm) supported on a lower polymer layer. Scale bar: $50\text{ }\mu\text{m}$. (A-III) Ultra-flexible nanoelectronic wire electrodes (NETs, cross-sectional area of $30\text{--}100\text{ }\mu\text{m}^2$, thickness of $1\text{ }\mu\text{m}$), achieved up to 18 modules with 128-channel longitudinal intracortex recording and 2304 simultaneous monitoring channels in behavioral mice and rats. Scale bar: $500\text{ }\mu\text{m}$. (B) Conformal-sheet design. (B-I) PEDOT: PSS or parylene C coated platinum and gold electrodes ($10 \times 10\text{ }\mu\text{m}^2$ electrode surface area and $30\text{ }\mu\text{m}$ interelectrode spacing). Scale bar: 1 mm . (B-II) High electrically conductive hydrogels (ECHs) electrode ($20\text{ }\mu\text{m}$ wide), dimethacrylate-functionalized perfluoropolyether (PFPE-DMA) as both the top and bottom elastomeric encapsulation layers. Scale bar: 1 mm . (B-III) Twining electrodes wrapped around the sciatic nerve (inner diameter of $\sim 2\text{ mm}$); polyimide ($2\text{ }\mu\text{m}$ thick) film with gold electrodes (200 nm thick) on a shape memory polymer ($\sim 100\text{ }\mu\text{m}$ thick) substrate. (C) 3D complex structure. (C-I) Electronics (platinum electrodes, each $\sim 1200\text{ }\mu\text{m}^2$ area, 50 nm thick) with mesh substrate (SU-8, $5\text{--}20\text{ }\mu\text{m}$ wide, $1\text{ }\mu\text{m}$ thick) were injected into $1 \times$ PBS solution with a glass needle. (C-II), Platinum electrode (diameter of $500\text{ }\mu\text{m}$ or $750\text{ }\mu\text{m}$) installed on a self-expanding Nitinol stent. Scale bar: $250\text{ }\mu\text{m}$. (C-III) 3D basket made of polyimide ($7\text{ }\mu\text{m}$ thick) in which eight gold electrodes are encapsulated (each $\sim 2000\text{ }\mu\text{m}^2$ in area and 300 nm thick). Scale bar: $400\text{ }\mu\text{m}$. Panels reproduced with permission from (A-I),^[23] Elon Musk, Neuralink; (A-II),^[62] Springer Nature Ltd; (A-III),^[63] Springer Nature Ltd; (B-I),^[25] Springer Nature Ltd; (B-II),^[64] Springer Nature Ltd; (B-III),^[65] AAAS; (C-I),^[29] Springer Nature Ltd; (C-II),^[66] Springer Nature Ltd; (C-III),^[67] PNAS.

controlling a certain thickness (20 μm).^[64] The aggregation of PEDOT polymers was promoted by blending ionic liquids into the conductive polymer (CP) solution of PEDOT:PSS. The ionic liquid additives are then removed by water exchange, followed by the transformation of the ionic gel to hydrogel; such ECH has very high conductivity (47.4 ± 1.2 S/cm). The direct contact of electrodes with the exposed sciatic nerve of anesthetized mice provided an excitation current density of up to 10 mA/cm² at a low voltage of 50 mV for neuromodulation.

The reduction of nerve damage through the implantation of unique self-deformable conformal-sheet electrodes is of great significance. The stretchable mesh serpentine wire is integrated into a flexible shape memory substrate (SMP) with permanent shape and reconfigurable and controllable elastic modulus (from ~ 100 MPa to ~ 300 kPa), which can naturally self-climb onto the nerve driven by 37°C normal saline, forming a flexible neural interface with minimal constraint on the deformable nerve (Figure 2B-III).^[65] The twinning electrode was implanted onto the rabbit sciatic nerve surface, and the results show a high signal-to-noise ratio (SNR) of 16 dB due to conformal contact with the nerve. The twinning electrodes proposed in this work have high potential for neuroelectrical stimulation and recording in clinical practice and basic neuroscience research.

2.3 | Implantable complex structures

Through the three-dimensional complex structure of flexible electrodes, direct interfacing between devices and the internal structures of man-made and biological materials can be further expanded and diversified.^[71] Figure 2C-I shows a submicron-thick (less than 1 μm), centimeter-scale (tens of square centimeters), macroporous mesh electronic component injected into artificial and biological cavities, or dense gels and tissues through syringes with needles as small as 100 μm in diameter.^[29] SU-8 is used as the coating material, with a total of 32 recording sites. The bending stiffness of the mesh electrode is as low as 8.7×10^{-11} N m², almost the same as that of the brain tissue. Stable electrical recordings of the mouse cortex for 8 months showed complete affinity between the reticular electrode and the brain tissue.

A long-term implanted platinum electrode array on a nitinol endovascular stent can be used to locally stimulate cortical tissue from within the vessel (Figure 2C-II).^[66] Six to twelve platinum electrodes with diameters of 500 μm or 750 μm were mounted on a nitinol self-expanding stent with super-elastic and shape-memory properties. These wires were spirally wound on stainless steel with a diameter of 310 μm and attached to a percutaneous connector implanted on the back of the animal. This minimally invasive electrode array was mounted on an endovascular stent for local stimulation of the sheep motor cortex. The findings have implications for understanding brain stimulation techniques and their potential therapeutic benefits.

Routes to three-dimensional mesostructures leveraging the nonlinear buckling of two-dimensional precursors, which are triggered by stress relaxation in prestrained elastomeric substrates, provide significant capabilities.^[67] The three-dimensional cages contain exposed circular gold pads (Figure 2C-III), which are exposed and serve as microelectrodes. The gold pads, measuring 50 μm in diameter and 300 nm in thickness, are coated with a 50 nm thick, biocompatible titanium nitride layer. They are specifically patterned to enable noninvasive extracellular stimulation and the recording of action potentials.

One drawback associated with flexible designs is that low bending stiffness makes it difficult for the flexible probes to penetrate the brain tissue, and therefore may cause more serious damage to the brain during insertion. Therefore, auxiliary reinforcing methods are required, such as the use of rigid guiding probes during insertion,^[61] injection through a syringe needle,^[71] flash freezing in liquid nitrogen,^[72] or coating the probe with soluble polyethylene glycol (PEG) reinforcer.^[73] Although significant progress has been made in the above systems, critical challenges remain for these emerging systems. For example, passive metal electrode materials (such as Au, Pt, etc.) do not have the same signal amplification and modulation function as active semiconductor transistors, and therefore the SNR of the acquired potential is low (<10 dB). Interfaces between conductive metal electrodes and brain tissue are prone to cerebrospinal fluid penetration, resulting in leakage current that may cause brain injury. These issues need to be carefully addressed when considering practical clinical applications.

2.4 | Interface optimization

The properties of the interface between the electrodes and the brain tissue need to be optimized when designing neural platforms. Brain signal recording with high SNR requires low impedance at the electrode-tissue interface. Generally, the impedance should be lower than 5 M Ω to avoid introducing high level of noise.^[74] For effective stimulation, electrodes should provide a high charge injection capability (CIC), the typical value of which in short excitation pulses (~ 1 ms) ranges from a few tenths of mC cm⁻² to several mC cm⁻².^[75] The electrical properties of commonly used electrode materials are summarized in Table 1; although they meet the basic requirement, they generally deteriorate when an insulation layer of glial sheath forms on the electrode's surface due to chronic immune response.^[76]

The impedance can be decreased by increasing the electrode size; however, this comes with a sacrifice in terms of the resolution and specificity of signal recording and stimulation. An alternative method is to extend the electrode surface area by using porous metal.^[77] For example, mesoporous metallic gold (MP-Au) film was synthesized by electrochemical deposition in Au electrolyte with micelles. An electrode (with a diameter of 60 μm) fabricated on a flexible PI substrate in this way offers a low impedance of

less than 1 k Ω at 1 kHz.^[78] In addition, electrode surface modification using conductive polymers, such as PEDOT, has been proposed due to its facilitation of high conductivity and effect in reducing foreign body response.^[79,80] A study has shown that a coating of tetrafluoroborate-doped PEDOT (PEDOT:BF₄) on platinum iridium (PtIr) electrodes extends the stable stimulation of neural probes implanted in mice to 60 days with an overall low impedance.^[81] To promote the long-term stability of PEDOT, roughening of electrode surfaces and the selection of different solvents and dopants have been investigated as methods for enhancing the bonding between coating layers and electrodes.^[82,83]

Hydrogel is another promising electrode coating material that has been used to act as interfacial media between neural electrodes and biological tissue, both providing improved electrical performance and enhancing the biocompatibility of the electrodes due to its water-rich feature and tissue-like mechanical properties.^[43,84,85] Conducting hydrogel coating on tungsten electrodes synthesized by the crosslinking of long-chain polymers (poly(styrene sulfonate-co-4-vinyl pyridine), Poly (SS-4VP)) and PEDOT has been reported to improve the stability of electrophysiological recording in freely moving mouse models.^[86] To enhance the adhesion between the hydrogel and the electrode, an aqueous-phase transfer printing strategy has been proposed for integrating multielectrode arrays (MEAs) on adhesive hydrogel-based substrates based on the in situ sol-gel phase transition of network precursors and hydrogel formation. The MEAs fabricated using this method possess ultracompliant mechanical properties (Young's modulus of 1–10 kPa) that match those of neural tissues.^[87]

3 | ACTIVE SEMICONDUCTOR-ENABLED NEURAL RECORDINGS

In active arrays, recording sites in each row and column share one data line and one scan line and can be controlled by multiplexing transistors, unlike in passive electrode arrays, where at least one wire connects with each electrode.^[88] Therefore, integration of active field-effect-transistor (FET) components into the neural electrode arrays as multiplexers and amplifiers can significantly decrease the number of wires required and provide per-channel signal amplification, leading to improved spatial resolution and SNR.^[89,90] In addition, neural interfaces with scalable recording arrays can be fabricated using micro-electromechanical system processing schemes that are compatible with CMOS technology, enabling high-resolution and large-scale recording.^[91] In the amplifier transistors of active interfaces, recording channels detect physiological potential through changes of its conductance; this has been achieved using inorganic/organic semiconductor materials including graphene, PEDOT:PSS, dinaphtho[2,3-b:2',3'-f]thieno[3,2-b]-thiophene (DNNT), pentacene, silicon, and so on.^[24,92–94]

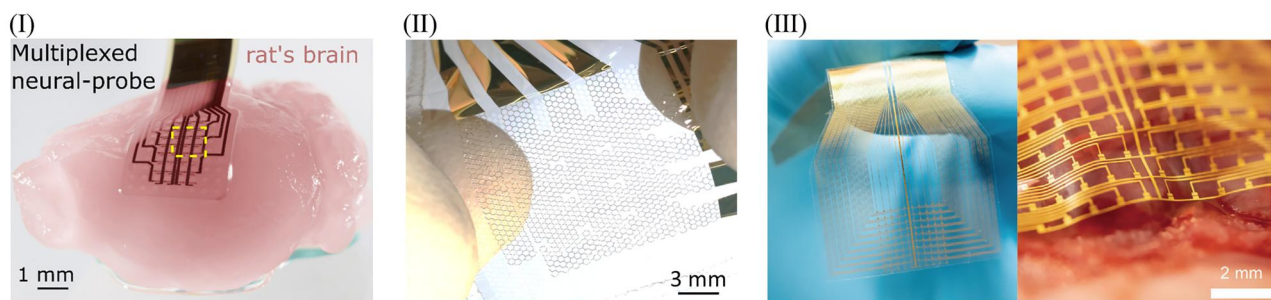
3.1 | Electrochemical semiconductor-based neural interfaces

Because of the outstanding properties of graphene in terms of flexibility, biocompatibility, and electrical mobility, graphene-based FET has been used in recording electrodes for developing neural interfaces in several reports.^[75,95] Figure 3A-I shows a flexible graphene transistor array fabricated on a polyimide substrate and encapsulated with SU-8.^[96] The recording array consists of 32 graphene solution-gated FETs (g-SGFETs) covering an area of $3.45 \times 4 \text{ mm}^2$ with 400 μm spacing between adjacent channels. The single-layer graphene channel (50 μm in width) prepared by chemical vapor deposition is connected to two metal leads, which serve as the source and drain terminals. When in contact with the brain tissue, the g-SGFET transduces the electrical potential variation through gate capacitance to the changes in the channel current. By using a frequency-division multiplexing (FDM) approach, this active sensor array avoids the use of on-site switches and enables a sampling rate up to 500 kHz. In vivo evaluation on rat brain validates the capability to measure visually evoked potentials and record cortical spreading depression (CSD) events.

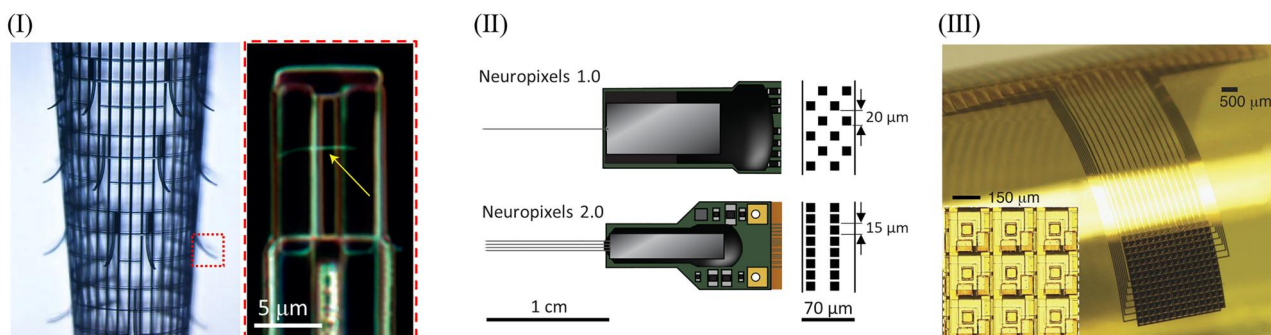
Conformal contact between the electrode interface and soft bio-tissues with complex surfaces requires high flexibility and stretchability of the recording devices. Figure 3A-II shows an ultrathin and stretchable multielectrode array with a matrix of active organic electrochemical transistors (OECTs).^[97] The 4×4 OECT array with PEDOT:PSS active channels ($10 \times 10 \mu\text{m}^2$) is fabricated on and encapsulated by 1.2- μm -thick deposited parylene layers, yielding an ultrathin sheet-like structure with a total thickness of 2.6 μm . The honeycomb grid design of the parylene substrate offers mechanical stability and structural stretchability. A coating of 100-nm-thick poly(3-methoxypropyl acrylate) (PM3A) as the outermost layer further increases the blood compatibility of the electrode array. One advantage of organic PEDOT:PSS is its high transconductance (typically in the order of millisiemen), which is two orders of magnitude higher than that of electrolyte-gated silicon transistors.^[102] Therefore, the OECT array is capable of performing ECG measurements on the heart surface of rats at an SNR of up to 52 dB. Such a stretchable OECT array with antithrombotic properties can be deployed on dynamically moving organs for long-term and stable signal monitoring.

Recent efforts have also focused on developing a biodegradable neural interface that completely disappears in the body through a bioresorption process after completing its functions.^[103,104] Figure 3A-III illustrates an example of a soft, bioresorbable OECT interface for the short-term monitoring of brain activity.^[98] Here, the OECT array consists of a PEDOT:PSS organic semiconductor, an Au electrode, and poly(lactic-co-glycolic acid) (PLGA) substrate, which have all been proven to be biocompatible materials.^[105–107] The 10×10 active PEDOT:PSS channels (rectangular area of $200 \times 20 \mu\text{m}^2$, 200 nm thick) distributed over an area of 0.64 cm^2 provide high transconductance up

(A) Electrochemical Semiconductor-based Neural Interfaces



(B) Silicon Nanostructured Devices



(C) Active Recording Mechanism (Capacitive vs Conductive)

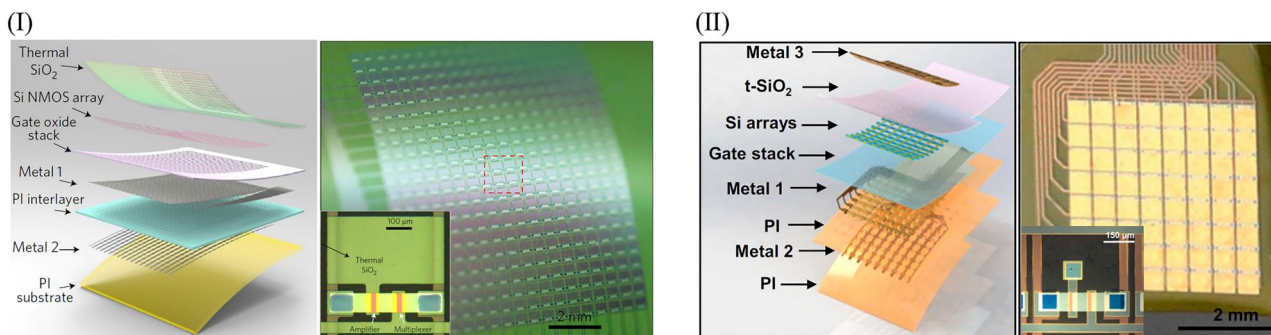


FIGURE 3 Active semiconductor-enabled neural recordings. (A) Electrochemical semiconductor-based neural interfaces. (A-I) Image of graphene solution-gated transistor array on a rat brain model for neural signal recording (32 recording sites with 400 μm spatial resolution, 50 μm graphene channel width). (A-II) OECT-based active recording array (16 sites, PEDOT:PSS active channel area of 10 × 10 μm²) encapsulated with a parylene layer (total thickness of 2.6 μm). (A-III) Left: bioresorbable transient OECT platform with 10 × 10 active PEDOT:PSS channels (channel area of 200 × 20 μm², 200 nm thick) with PLGA substrate and insulating layer (12 μm/100 nm thick). Right: image of the thin, flexible OECT platform attaching to a rat's brain. (B) Silicon nanostructured devices. (B-I) Left: ultra-flexible macroporous cylindrical probe (diameter of 100–200 μm) assembled from 2D mesh structures with 19 Si NW transistors on the supporting arms distributed on the surface. Right: Enlarged image of the Si NW transistor with metal contact (Cr/Pd/Cr, 1.5/50–80/1.5 nm thick). (B-II) Schematic of the Neuropixels platforms with high-density multiplexed titanium nitride electrodes on the silicon shank (1000 electrodes on one shank in Neuropixels 1.0, 1280 electrodes on each of the four shanks in Neuropixels 2.0). (B-III) Image of a μ-ECoG platform with actively multiplexed Si NM transistors (14 × 14 array, 250 μm spatial resolution, 260 nm thick) fabricated on a flexible polyimide substrate (1.2 μm thick) and encapsulated by polyimide/epoxy layers (1.2/4 μm thick). Inset shows an enlarged view of a 3 × 3 region of the transistor array. (C) Active recording mechanism (capacitive vs. conductive). (C-I) Left: exploded-view schematic illustration of a high-channel-count active μ-ECoG array of capacitively coupled Si NM transistors encapsulated by t-SiO₂ layers (900 nm thick). Right: photograph of the 18 × 22 transistor array (coverage area of 9.5 × 11.5 mm²). Inset shows a unit recording cell with a multiplexer transistor and a capacitively coupled amplifier transistor (channel length of 13.8 μm, width of 80 μm, thickness of 50 nm). (C-II) Left: schematic illustration of an active μ-ECoG array of conductively coupled Si NM transistors encapsulated by a t-SiO₂ layer (1 μm thick) and a PI sheet (Kapton, 13 μm thick). Right: photograph of the 8 × 8 transistor array (coverage area of 3.2 × 3.2 mm²). Inset shows a unit recording cell with a multiplexer transistor and a conductively coupled amplifier transistor (channel length of 16 μm, width of 80 μm, thickness of 60 nm). Panels reproduced with permission from (A-I),^[96] American Chemical Society; (A-II),^[97] AAAS; (A-III),^[98] Wiley; (B-I),^[72] Springer Nature Ltd; (B-II),^[99] AAAS; (B-III),^[100] American Physiological Society; (C-I),^[51] Springer Nature Ltd; (C-II),^[101] PNAS.

to 8.67 mS. The results show that such an OECT implant can be bent with a radius less than 5 mm without significant variation in transconductance. In vivo experiments on the cortical surface of rats within six days demonstrated the capability to achieve real-time monitoring of electrical activities at high SNR (37 dB) and spatiotemporal resolution (1.42 ms and 20 μm). Such devices can provide continuous and temporary recording and then dissolve at prescribed times, eliminating the necessity of a secondary extraction surgery and the associated risk and cost.

3.2 | Silicon nanostructured devices for neural recordings

Neural probes designed with low stiffness and ultraflexibility to match the property of brain tissue represent another direction of research in this field.^[31] Figure 3B-I demonstrates a microporous brain probe that combines high flexibility (bending stiffness $<0.64 \times 10^{-15} \text{ N m}^2$) and subcellular feature size ($\sim 10 \mu\text{m}$).^[72] This three-dimensional cylindrical probe (100–200 μm in diameter) integrated with 19 silicon nano-wire FETs is self-assembled from two-dimensional mesh structures (7 μm wide SU-8 parallel ribbons with a thickness of 300–500 nm as the bottom and top passivation layers) on a silicon substrate with a sacrificial layer (100 nm thick Ni). After dissolving the sacrificial layer, the transverse elements (3 μm wide Cr/Pd/Cr metal film ribbons with thickness of 10–20/80/1.5 nm) with built-in compressive strain shape the mesh into a cylindrical global structure, while the FETs bend outward from the probe along with the supporting arms (Cr/Pd/Cr metal ribbons with thickness of 1.5/80/50–80 nm), serving as the voltage sensors around the probe. The force required to bend the supporting arms by 10 μm is only about 10 nN, which is on the same level as the migration force of single cell.^[108] Insertion of such a frozen-state probe into rodent brains and successful recording of multiplexed local field potentials and single-unit action potentials demonstrates its capability for low-damage neural recording.

Integration with the complementary CMOS technology enables the development of neural interfaces with reduced size and much higher electrode density. Figure 3B-II shows Neuropixels platforms as an example. Neuropixels 1.0 features a penetrating probe with nearly 1000 multiplexed titanium nitride electrodes (area of $12 \times 12 \mu\text{m}^2$, 15 μm spacing) along a rigid silicon shank (10 mm long, $70 \times 20 \mu\text{m}^2$ cross section) and integrated CMOS circuits for addressing, filtering and amplification.^[22] The updated four-shank version of the probe (250 μm center-to-center spacing between shanks) in Neuropixels 2.0 further increases the electrode density by minimizing the spacing between electrodes to 15 μm , providing 1280 recording sites on each shank.^[99] The high-count electrodes of the Neuropixels probes allow both the stable activity recording of multiple neurons simultaneously from different brain regions of freely moving rats for months after implantation, and the acute recording of neuronal activity in a human cortex in an operating room environment.^[109–111]

Neural recording with penetrating probes captures more accurate signals in the deep brain due to direct interaction with neurons but comes with inherent shortcomings in terms of invasiveness because the probes may damage blood vessels as they are inserted into the cerebral cortex.^[112] Alternative platforms which alleviate the invasiveness of penetrating probes, but also utilize the extremely high channel count and scalability of silicon active devices, monitor ECoG from the cortex surface and the subgaleal space with a larger area in a less invasive manner.^[103,113] ECoG has been proven to provide key information for elucidating the fundamental pathogenesis of various neurological disorders, including epilepsy and Parkinson's disease.^[9,114] Figure 3B-III shows a notable example of an active $\mu\text{-ECoG}$ array with integrated electronics fabricated on a flexible polyimide substrate and encapsulated by polymetric layers (overall thickness $\sim 25 \mu\text{m}$).^[100] Here, the $\mu\text{-ECoG}$ recording array contains 196 platinum electrodes (50 nm thick) arranged into a 14×14 matrix with 50 μm spacing between adjacent recording sites and a total coverage area of $3.5 \times 3.5 \text{ mm}^2$. Each site uses two transistors in the transfer-printed U-shaped silicon nanomembrane ($\sim 260 \text{ nm}$ thick), one connected with the electrode for buffering the ECoG signal and the other for active addressing. This active multiplexing circuit design requires only 29 interface wires in the 196-electrodes array and yields a sampling rate of $\sim 297.6 \text{ Hz}$ per electrode. The lifetime of such a recording array can be further extended with effective coating and encapsulation materials.

3.3 | Capacitive and conductive recording mechanisms

In multiplexed active recording arrays, the transistors acting as voltage amplifiers detect potential variation through the capacitive or conductive coupling between the contacting tissue and the sensing transistors. Figure 3C-I shows a 396-channel multilayer-structured active matrix with capacitively coupled Si NM transistors in an 18×22 array layout and with a total area of $9.5 \times 11.5 \text{ mm}^2$.^[51] Here, the 900-nm thick thermally grown SiO_2 (referred to as t- SiO_2) serves as an encapsulation layer for the underlying electronics as well as the dielectric medium, giving rise to capacitive coupling between the tissue and the metal gate (Cr/Au, 5/100 nm, area of $270 \times 460 \mu\text{m}^2$) of the Si NM transistor for voltage signal amplification. The results of a soaking test, which involved the hydrolysis of t- SiO_2 in PBS solution (pH 7.4) at 37°C , demonstrate a low leakage current, lower than $10^{-9} \text{ A cm}^{-2}$ within 120 days, indicating the excellent encapsulation performance of t- SiO_2 . This capacitive sensing method requires a sufficiently large gate pad area and a sufficiently thin t- SiO_2 layer to ensure a much higher electrode-tissue capacitance than that of the gate, imposing a trade-off between channel density, device lifetime, and amplification efficiency. An alternative conductive approach shown in Figure 3C-II comprises an 8×8 active matrix with a multiplexed



addressing transistor and a conductively coupled transistor at each unit site ($360 \times 360 \mu\text{m}^2$).^[101] An opening with a diameter of $100 \mu\text{m}$ through the $1 \mu\text{m}$ thick t-SiO₂ encapsulation layer to the highly doped 60 nm thick P⁺⁺-Si layer defines an electrical contact between the metal gate of the Si NM transistor and the Au pad deposited on the back of the device. This dual-side configuration of P⁺⁺-Si and t-SiO₂ forms a bilayer barrier in the device and makes the Au pad act as a conductively coupled biointerface for recording or stimulation. A soaking test in 96°C PBS yielded stable operation for 30 h, suggesting a projected device lifetime of 285 days at physiological temperature (37°C). Compared with the capacitive method, the conductive approach can provide high-fidelity recording and efficient stimulation due to the conductive coupling between the electrode and the targeted tissue.

4 | ADVANCED APPROACHES FOR CHRONICALLY STABLE SYSTEM

4.1 | Active, high channel-count systems for long-term bio-integration

In neural interfaces where metal electrodes are in direct contact with the biological tissue, the leakage current generated by the biofluid penetration increases the risk of device failure and damage to neural tissues. Therefore, effective encapsulation materials are required in the active interfaces intended for long-term implant applications. As mentioned above, thin t-SiO₂ film has been proven to act as an effective barrier for active array platforms due to its defect-free and low water penetration properties.^[115] Figure 4A-I shows a kilo-scale, high-density active matrix (Neural Matrix) with $1 \mu\text{m}$ thick t-SiO₂ layers serving as both biofluid barriers (front and back sides) and capacitive coupling interfaces (front side).^[26] The device integrates 1008 capacitively coupled electrodes ($195 \times 270 \mu\text{m}^2$ area, $60\text{-}\mu\text{m}$ spacing) arranged into a 28×36 matrix with a total area of $9 \times 9.24 \text{ mm}^2$, with only 92 interconnecting wires by taking advantage of the multiplexing transistors (Figure 4A-II). Such a thin and highly flexible Neural Matrix ($29\text{-}\mu\text{m}$ thick in sensing area) maintained nearly 100% yield and negligible leakage current during repeated bending tests (10,000 cycles, minimum bend radius of 2.5 mm). Chronically implanted on the auditory cortex of rats, the array successfully detected the evoked responses over a period of 435 days, with a leakage current well below the safety threshold of $1 \mu\text{A}$ (Figure 4A-III). Spatiotemporal pattern recording of visual-evoked potentials moving across sensorimotor cortices in nonhuman primates (NHPs) shown in Figure 4A-IV demonstrates the long-term, high-definition mapping capability of the Neural Matrix arrays.

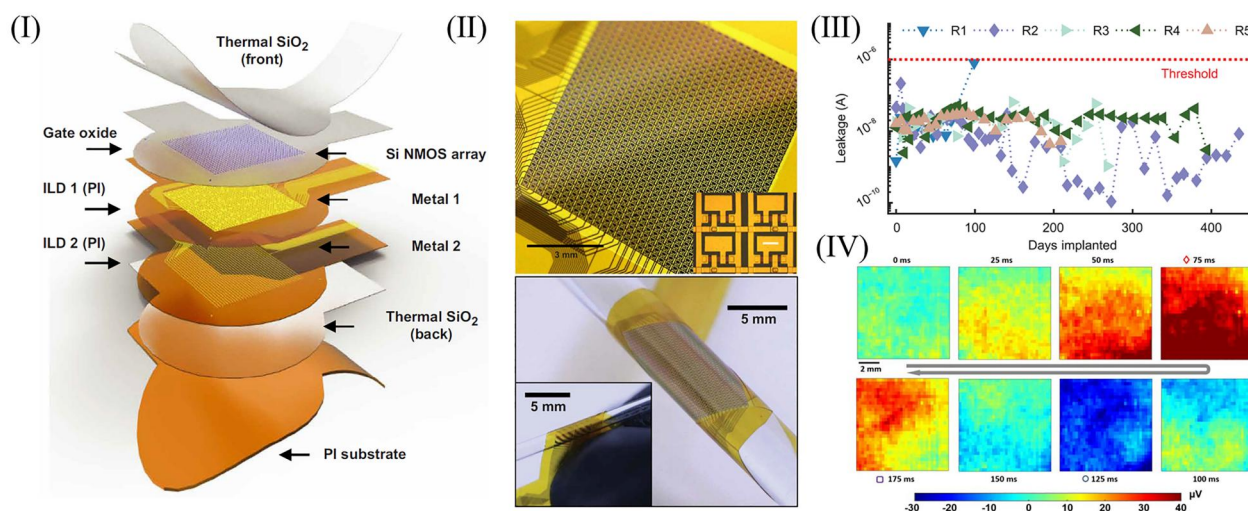
4.2 | High scalability bioelectronics with variable densities

Developing high-scalability, long-lived, and shape-conformal electrical interfaces with layouts customized to the size and functional heterogeneity of large brain models represents an attractive direction of current research. Figure 4B-I illustrates an advanced approach for this purpose using micro-scale transfer-printing techniques.^[118,119] Specifically, tens of thousands of microscale device components (Si NM transistors and inorganic light emitting diodes) derived from wafer sources are transfer-printed on shape-conformal polymer film using patterned PDMS stamps with various densities and pitch spacings, resulting in interconnected functional networks.^[116] The example in Figure 4B-II shows more than 32,000 microdie printed on a flexible polyethylene terephthalate (PET) sheet with an outline and covering area ($\sim 150 \text{ cm}^2$) comparable to that of an adult human brain model. Variable density design in the microdie array ($350 \times 440 \mu\text{m}$ spacing in sparse areas, $100 \mu\text{m} \times 110 \mu\text{m}$ spacing in dense areas) targets the neural recording of different brain functional regions (i.e., motor and sensory regions). An immersion test of a t-SiO₂ ($1 \mu\text{m}$ and 900 nm on two sides) encapsulated array in 96°C PBS (pH 7.4) yielded stable operation throughout ~ 9 days, which can be projected to a device lifetime of over 60 years at physiological temperature with Arrhenius scaling.

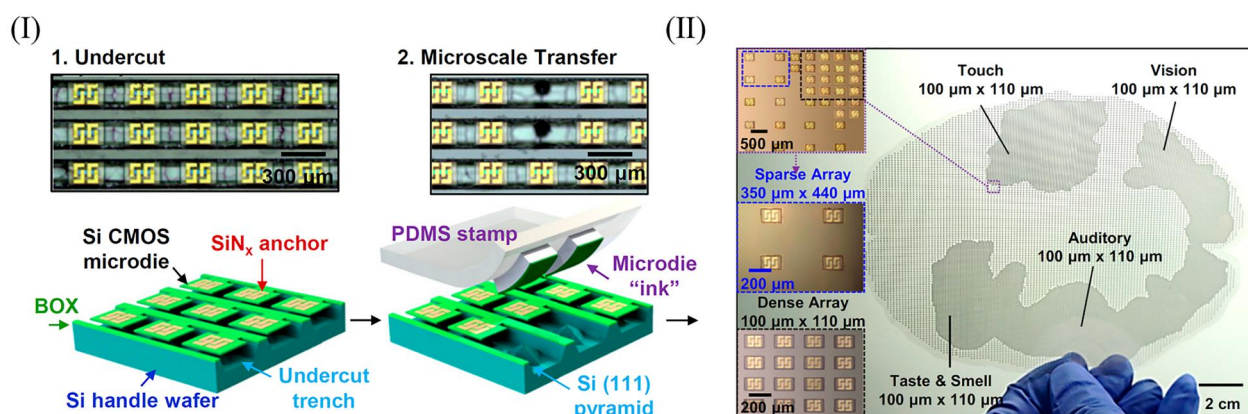
4.3 | Long-lived neural recoding/ stimulation with SiC interfaces

Continuing efforts are being devoted to extending the channel count, resolution, signal quality, and life span of neural recording interfaces, as summarized in Table 2. Long-lived flexible neural interfaces that are capable of simultaneous stimulation and recording are of broad interest to neuroscience research and therapy for neurological disorders. Silicon carbide (SiC) has emerged as a promising platform for long-term electronic biointerfacing. Compared with commonly used metal materials, SiC provides additional advantages, including impedance which can be adjusted by doping, excellent biocompatibility, extremely low hydrolysis rate, and chemical inertness.^[117,127] Figure 4C-I illustrates a flexible neural electrode with a multilayered structure on a polyimide substrate ($10 \mu\text{m}$ thick), where the highly doped SiC nanomembrane (200 nm thick) and t-SiO₂ thin film (600 nm thick) serve as the faradic interface and electrical barrier, respectively.^[128] In vivo nerve stimulation performed on a rat (Figure 4C-II) has demonstrated the potential application of such a SiC/SiO₂ platform to neuronal stimulation and the restoration of body function.

(A) Active, High Channel-Count Systems for Long-Term Bio-integration



(B) High Scalability Bioelectronics with Variable Densities



(C) Long-Lived Neural Recoding/Stimulation with SiC Interfaces

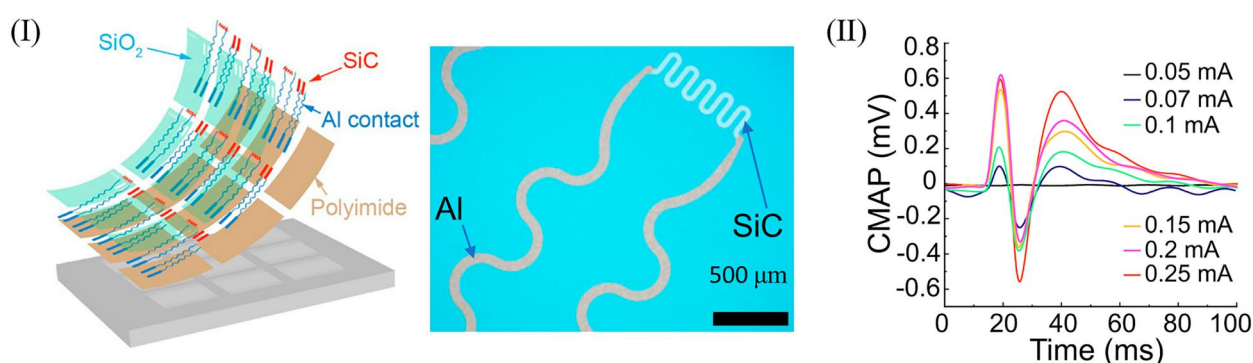


FIGURE 4 Advanced approaches for chronically stable neural interface platforms. (A) Active, high channel-count systems for long-term bio-integration. (A-I) Exploded view of the high-definition active μ -ECOG neural interface (Neural Matrix) with t-SiO₂ encapsulation layers on both sides (t-SiO₂ layer thickness of 1 μ m). (A-II) Above photograph of the 28 $\times$ 36 capacitively coupled active transistor array (coverage area of 9 $\times$ 9.24 mm²). Inset shows an enlarged view of a 2 $\times$ 2 cell of the transistor array (cell area of 330 $\times$ 250 μ m²). Below: photograph of the electrode array bent around cylindrical tubes with different diameters (diameter of 5 mm in the main image and 2.5 mm in the inset). (A-III) Leakage current of the Neural Matrix interface implanted on a rat cortex over 435 days. (A-IV) Spatiotemporal maps of visual-evoked potentials of the NHPs sensorimotor cortices recorded by the Neural Matrix platform. (B) High scalability bioelectronics with variable densities. (B-I) Schematic illustration of the transfer printing of silicon microdie from source wafer. (B-II) Photograph of ~32,000 transfer-printed microdie array in variable densities on a flexible polymer film with outline and size approximate to that of human brain. (C) Long-lived neural recoding/stimulation with SiC interfaces. (C-I) Left: Exploded view of the flexible long-lived SiC/SiO₂ bioelectronic interface on polyimide substrate for neural recording and stimulation (PI/SiC/SiO₂, 10 μ m/200 nm/600 nm thick). Right: enlarged image of the serpentine SiC electrode connected with Al interconnects. (C-II) Recorded amplitude of compound motor action potential (CMAP) for a rat sciatic nerve at different stimulating currents from the SiC electrode interface. Panels reproduced with permission from (A),^[26] AAAS; (B),^[116] PNAS; (C),^[117] PNAS.



TABLE 2 Comparison of key performance of recent neural interfaces.

Channel counts	Spacing (mm)	Coverage (mm ²)	Recording mechanism	SNR (dB)	Lifetime (days)	Subject	Ref.
64	0.1	1	Passive	-	10	Rat	[25]
128	0.15 × 0.06	~1.15	Passive	5.51	290	Mouse	[63]
64	0.4	10.24	Active	42	33	Rat	[103]
100	0.57 × 0.64	64	Active	37	6	Rat	[98]
1280	0.032 × 0.015	~0.7	Active	-	250 (rat)	Rat/Human	[99]
196	0.25	12.25	Active	-	-	Rat	[100]
1008	0.33 × 0.25	83.16	Active	-	-	NHP	[26]
32	0.5	4	Passive	32.8	20	Mouse	[120]
32	0.2	0.84	Passive	-	91	Rat	[121]
64	0.4	12	Active	-	6	Rat	[122]
72	0.64	27	Passive	-	42	Rat	[123]
240	0.23	840	Passive	-	-	Human	[124]
252	2	2100	Passive	-	137	NHP	[125]
1024	1	2014	Passive	-	-	Rat/Human	[28]
1152	0.295	98	Passive	-	-	NHP	[126]

5 | CHALLENGES AND PERSPECTIVES

As demonstrated in this paper, research on advanced materials, innovations in structural design, and improvements in integration schemes has laid the foundation for the development of flexible bioelectronic systems with long-term stability and excellent performance. Many kinds of materials, such as metals and electrically conductive hydrogels, are used in research and development.^[23,64] Commonly used active materials usually have a low channel count (i.e., in the hundreds of channels), a small acquisition area, and yet remain rigid (Young's modulus of GPa). As a solution, silicon-based materials (such as Si, SiO₂, SiC, and silicide) that can provide both active electronic functions and long-term stable operation are the focus of current research.^[22,99,101,129] They can be multiplexed to achieve fewer transistors and more channels and can amplify signals to obtain weak signals. In addition, recent studies have shown that using thin, transfer-printed SiO₂ layers with different thicknesses produced by thermal growth on silicon wafers as barrier layers can well adjust encapsulation time up to 60 years.^[116]

Simply changing the nature of the electrode from rigid to flexible is a long way from meeting the current needs of flexible bioelectronic system applications, particularly in complicated in vivo surrounding environments. A key feature for high performance neural electronic interfaces is the ability to meet biological compatibility (shape conformability^[130] and stretchability^[131]), high spatial and temporal resolution for high-fidelity operation, and scalability (thousands of active channels) requirements.^[32] Due to the soft, elastic, variable, and reconfigurable

characteristics of biological tissues, it is particularly important that neural interfaces continue to work in bent, stretched, and twisted states. A common solution is to combine the advantages of different materials (organic or inorganic) in an integrated approach to form the most suitable system.^[132] In addition, the miniaturization of devices is among the effective ways to reduce negative effects, but inevitably increases system impedance and reduces signal quality. High-density data acquisition by miniaturized arrays also requires a dedicated data acquisition system, which often limits its application. Although meeting these requirements represents a significant challenge, the potential applications and future prospects of the device are extremely attractive.

The emerging chronic neural interface systems highlighted in this paper play important roles in basic research on cell cultures, organoids, and animal models.^[42,44,46] However, long-term biocompatibility and ethics are crucial for future practical applications, and the devices currently being studied have only been used on animals such as mice, rats, rabbits, and pigs, as shown in Table 2. Although many theoretical systems have yielded promising results in scientific laboratories, the long-term effects of these materials on the human body are still unpredictable, which limits their commercial use in humans. Even those approved by the FDA, such as the Utah array, are made of rigid silicon and metals that may cause damage to the human brain and are not suitable for long-term applications. In fact, an increasing number of devices are being approved for human applications, such as the Neuralink and endovascular Stentrode devices discussed in this paper. The innovation represented by these bioelectronic systems is expected to have an

enormous impact on promoting the development of neural interfaces for human applications.

AUTHOR CONTRIBUTIONS

Lianjie Zhou: Investigation; Visualization; Writing—original draft. **Zhongyuan Wu:** Investigation; Visualization; Writing—original draft. **Mubai Sun:** Investigation; Data curation. **Jaemin Park:** Investigation; Data curation. **Mengdi Han:** Validation; Resources. **Ming Wang:** Validation; Resources. **Junsheng Yu:** Validation; Resources. **Zengfeng Di:** Validation; Resources. **Yongfeng Mei:** Validation; Resources. **Wubin Bai:** Methodology; Validation; Resources. **Xinge Yu:** Methodology; Validation; Resources. **Ki Jun Yu:** Methodology; Validation; Resources. **Enming Song:** Conceptualization; Methodology; Resources; Supervision; Writing—review & editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare that they have no conflicts of interest.

DATA AVAILABILITY STATEMENT

All data used to support the findings of this study are included in the article. Raw data used to generate the figures are available from the corresponding author upon request.

ETHICS STATEMENT

Ethics approval was not needed in this study.

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